Gemce

AI study buddy that works anywhere you go.

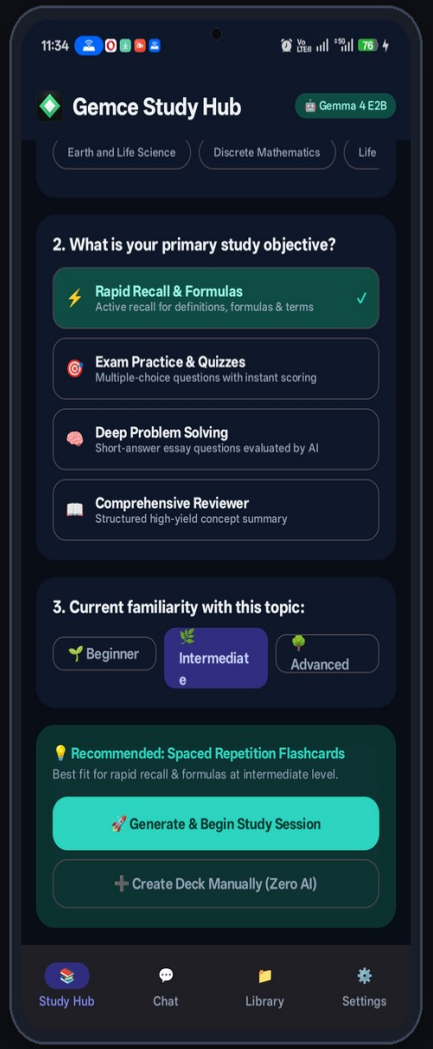
The 100% offline, privacy-first Android study platform powered by on-device Google Gemma 4 E2B via LiteRT-LM. Eliminating the connectivity divide in education.

🔒 100% Offline & Private

⚡ Gemma 4 E2B via LiteRT

📚 Verbatim Source Citations

🧠 SuperMemo SM-2



01 · THE PROBLEM & CONTEXT

Millions of Students Cannot Rely on Cloud AI

Cloud-based learning platforms collapse the moment connectivity drops.

📶 Connectivity & Latency Barriers

- High data tariffs and spotty mobile reception across rural schools, transit commutes, and dormitories.
- Cloud AI tools (ChatGPT, Claude, Gemini Web) completely fail when data runs out or signal is lost.
- Students are cut off from AI study tools right when they need exam revision the most.

💳 Prohibitive Subscription Paywalls

- $20/month subscription fees are unattainable for mainstream students in developing economies.
- Proprietary paywalls gatekeep high-quality tutoring behind credit cards and recurring payments.
- Educational inequality widens as only connected students access AI assistance.

🛡️ Data Privacy & Cloud Surveillance

- Uploading student lecture notes, essays, and study materials creates unmonitored cloud data harvesting.
- No data sovereignty: student materials are retained on external corporate servers.
- Institutional compliance issues when handling proprietary textbooks or curriculum syllabi.

💡 **Fundamental Principle: Quality academic assistance should never be contingent on having a cellular data connection.**

02 · THE SOLUTION & ARCHITECTURE

Gemce: On-Device AI Study Architecture

Sovereign on-device intelligence executing Google's Gemma 4 E2B via LiteRT-LM.

🔒 100% Local Inference & Offline Guarantee

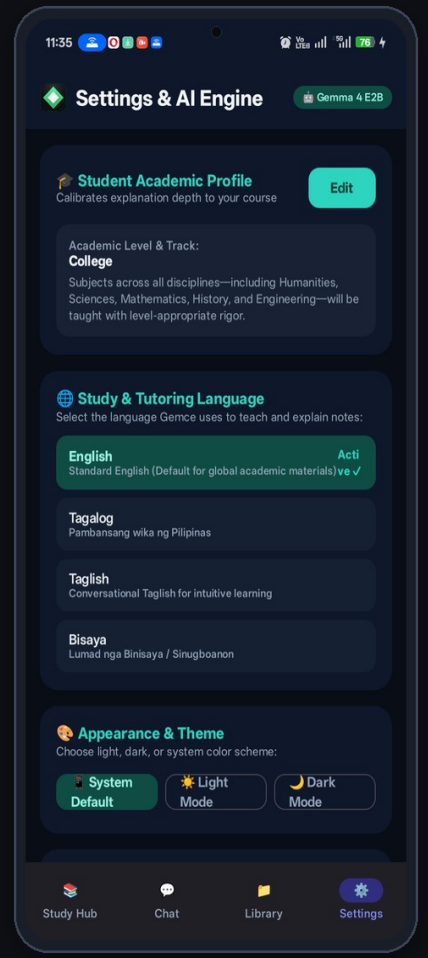
- Over 2.4 GB of quantized model weights stored permanently in on-device storage.
- Zero outbound network requests during study sessions. No tracking, telemetry, or cloud dependencies.

⚡ Google LiteRT-LM GPU Accelerated Engine

- Direct execution on Android OpenCL GPU backend (Adreno / Mali).
- Sub-second Time-To-First-Token and smooth streaming token generation on modern Android devices.

📦 Persistent Weights & Zero-AI Fallback

- Model file survives app updates, cache clears, and uninstalls.
- Manual flashcards, deck browsing, and SM-2 scheduling remain 100% functional even with zero AI.



03 · CURRICULUM CALIBRATION

Tailored Rigor Across Disciplines & Languages

Calibrates explanation depth, curriculum context, and vocabulary to the student.

🎓 Dynamic Academic Level Calibration

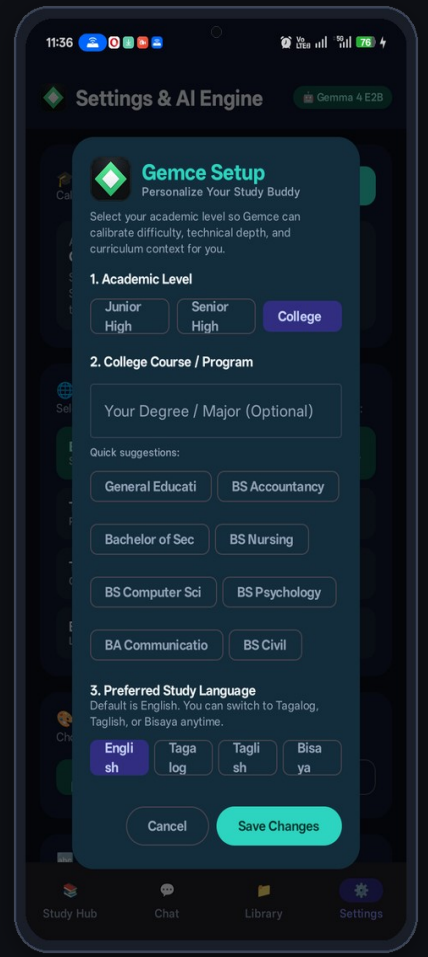
- Calibrates from Junior High, Senior High tracks (GAS, HUMSS, ABM, STEM, TVL, Arts) to College majors.
- Aligns technical vocabulary and pedagogical depth to university programs like Accountancy, Nursing, Education, and Engineering.

🌐 Multilingual Academic Tutoring

- Speaks English, Filipino (Tagalog), Taglish, and Binisaya (Cebuano).
- Students understand tough conceptual mechanisms in their natural, conversational tongue.

📖 Packaged Reading Typography

- Bundles packaged TrueType fonts: Calibri (modern), Serif (editorial), and Monospace (STEM).
- System-wide font selection optimizes legibility for dense textbook reading sessions.



04 · STUDY METHODOLOGY

Adaptive Study Hub & Intelligent Diagnosis

Transforms raw syllabus topics and lecture notes into goal-oriented learning workflows.

⚡ 1. Rapid Recall & Formulas

- Diagnoses active retention goals.
- Prescribes Spaced Repetition Flashcards powered by SuperMemo SM-2.

🎯 2. Exam Practice & Quizzes

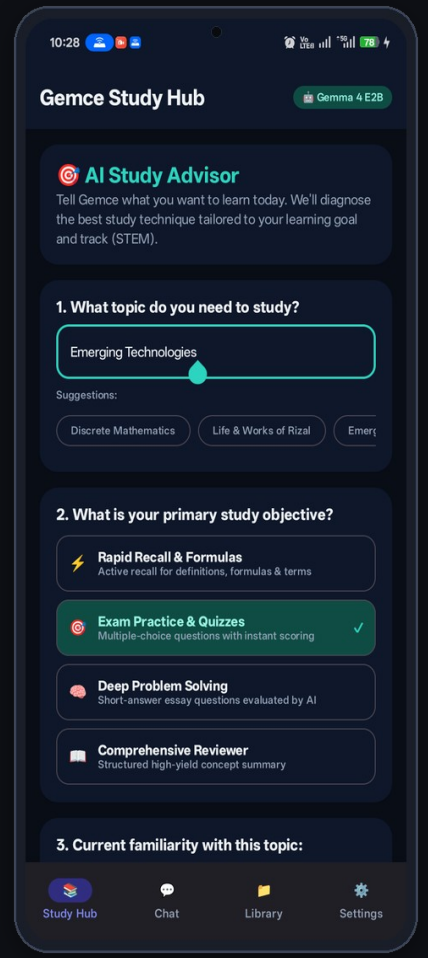
- Diagnoses test preparation goals.
- Prescribes instant Multiple-Choice Questions with rationale feedback.

🧠 3. Deep Problem Solving

- Diagnoses deep conceptual synthesis.
- Prescribes short-answer essay questions evaluated against rubrics.

📖 4. Comprehensive Reviewer

- Diagnoses structured pre-exam cramming.
- Prescribes high-yield bulleted concept summaries with source quotes.



05 · ACTIVE RECALL & RETENTION

Evidence-Grounded Flashcards with SuperMemo SM-2

Combines the gold-standard spaced repetition algorithm with zero-hallucination provenance.

🧠 SuperMemo SM-2 Spaced Repetition

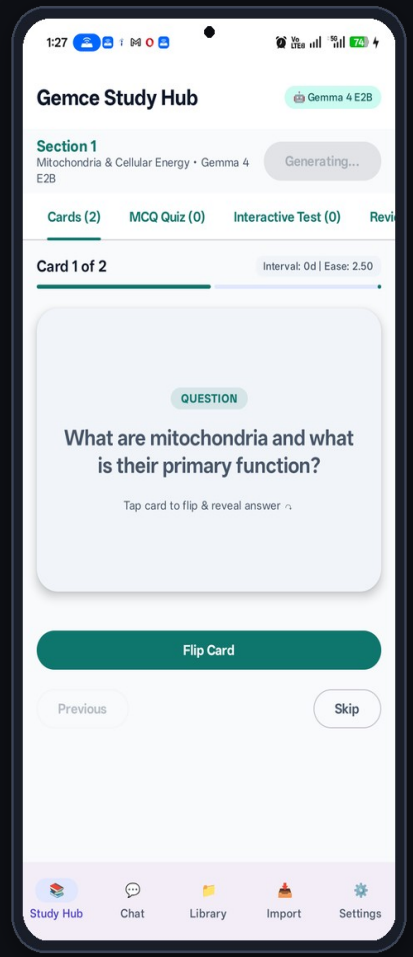
- Automatically calculates ease factor (EF) and repetition intervals (I_n).
- Self-evaluations: Again (1d), Hard (3d), Good (4d), Easy (6d) optimize permanent retention.

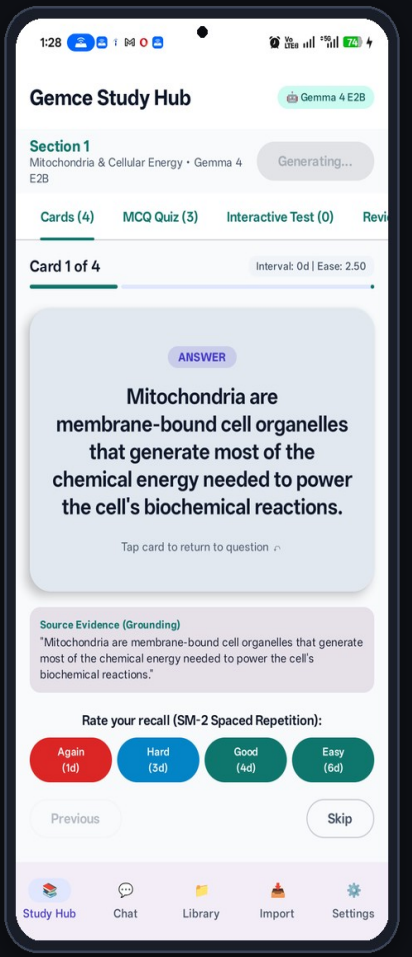
🚫 Strict Negative Prompt Constraints

- Explicitly forbids generic meta-questions like 'What was the user intended to study?'.
- Rigorously tests concrete academic terms: Mitochondria, ATP synthesis, cellular respiration.

📖 Verbatim Evidence & Source Grounding

- Card back highlights verbatim source sentences and section provenance tags.
- Guarantees 100% accurate studying without fabricated AI hallucinations.





06 · EXAM PREPARATION

High-Yield Reviewers & Verified Practice Quizzes

Synthesizes complex notes into structured study sheets and scored assessments.

📝 High-Yield Topic Synthesis

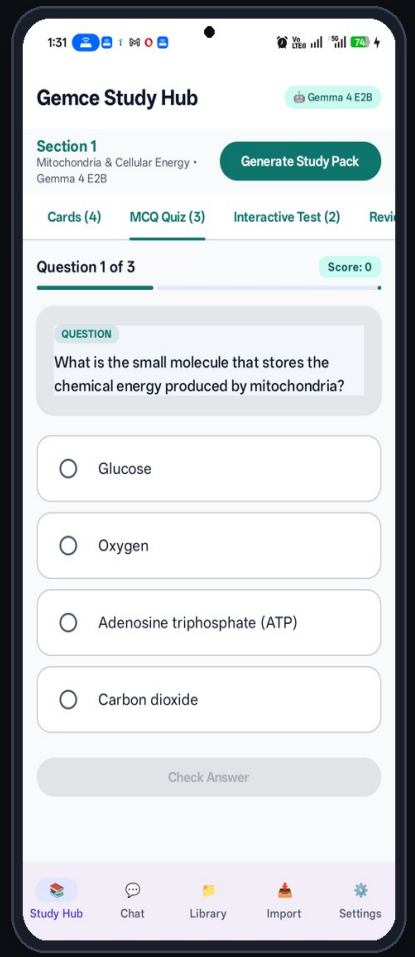
- Organizes dense textbook chapters into bulleted key principles, classifications, and definitions.
- Ideal for rapid revision before major prelim, midterm, and board examinations.

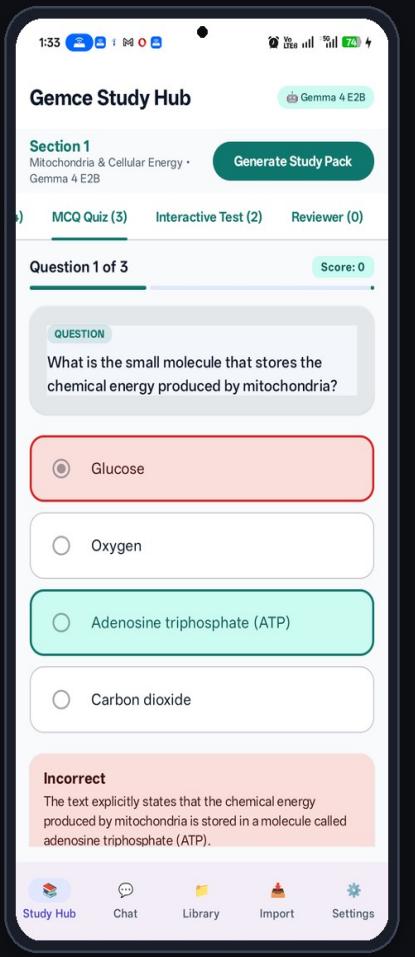
🏷️ Provenance & Verifiability Badges

- Cites exact source sentences for every generated key concept.
- External facts added by the model are visibly marked [unverified] for complete transparency.

🎯 Multiple-Choice Exam Practice

- Generates rigorous 4-option MCQs testing conceptual understanding.
- Instant grading with detailed explanations on why distractor options are incorrect.





07 · INTERACTIVE TUTORING

Real-Time Conceptual Q&A in Airplane Mode

Interactive academic mentor explaining tough mechanisms whenever curiosity strikes.

🌍 Universal Multidisciplinary Grounding

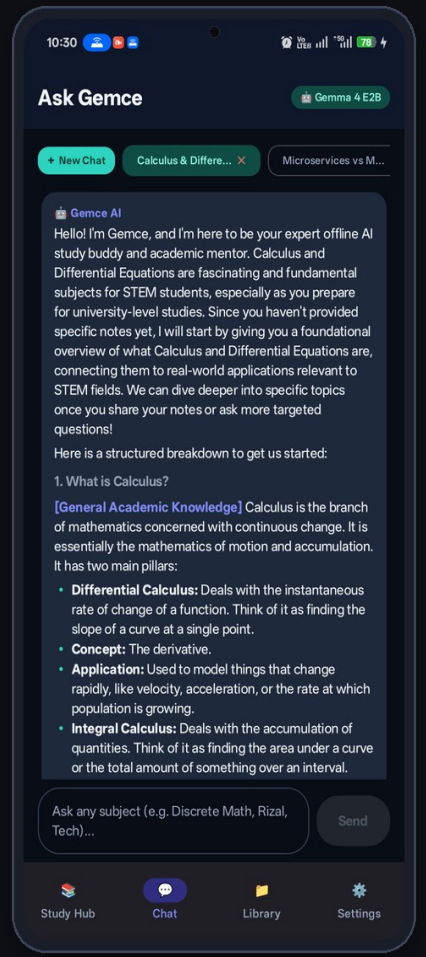
- Mentors across Natural Sciences, Discrete Mathematics, World History, Rizal, Law, and Medicine.
- Resolves complex queries: e.g. rock cycle mechanisms, truth tables, and historical contexts.

📑 Rich Markdown & Code Formatting

- Native Jetpack Compose Markdown rendering with formatted tables, lists, and KaTeX math formulas.
- Clean visual hierarchy avoids walls of unstructured text.

💾 Multi-Session Local Memory & Tabs

- Conversation history persisted securely on device with quick topic tabs.
- Zero cloud leakage: student inquiries never leave device memory.



08 · ENGINEERING & TECH STACK

Engineered for Native Android Performance

Clean MVVM architecture built with modern Kotlin and on-device machine learning.

🎨 UI & Presentation Layer

- 100% Jetpack Compose + Material 3.
- Dark / Light / System theme switching.
- Edge-to-edge support across Android 14+.
- Integrated TrueType fonts (Calibri, Serif, Mono).

🤖 On-Device AI Engine

- Google LiteRT-LM runtime engine.
- Quantized Gemma 4 E2B weights.
- OpenCL / Adreno GPU acceleration.
- Decoupled LlmEngine with test FakeEngine.

📥 Document Ingestion Pipeline

- Offline Google ML Kit Text Recognition v2.
- Pure local PDF extraction via pdfbox-android.
- Normalizes photos, PDF, DOCX, and text paste.
- Clean section parsing with verbatim linking.

💾 Local Persistence & Reliability

- Android Room Database + SharedPreferences.
- WorkManager for background model tasks.
- Model stored in persistent app storage.
- Zero network calls beyond weight download.

09 · ROADMAP & CALL TO ACTION

Democratizing AI for Every Student Everywhere

Gemce is open-source, fully verified on physical hardware, and ready for deployment.

✅ Shipped MVP Milestones (v0.2.2)

- Full compliance with PLAN.md architecture: OCR/PDF ingest, SM-2 flashcards, MCQs, Reviewers, and Ask Gemce chat.
- Verified on Android 14 (OPPO Reno12 F 5G) with Google Gemma 4 E2B model weights.

🚀 Immediate Roadmap (v0.3)

- Anki Deck (.apkg) bidirectional export & import.
- Audio flashcard review mode (hands-free offline playback).
- Camera instant reviewer scanner with camera auto-cropping.

📲 Open Source & Distribution

- GitHub Repository: github.com/meloervy/appbuild.gemce
- Installable APK: Packaged via GitHub Actions CI/CD (Gemce-v0.2.2.apk).

